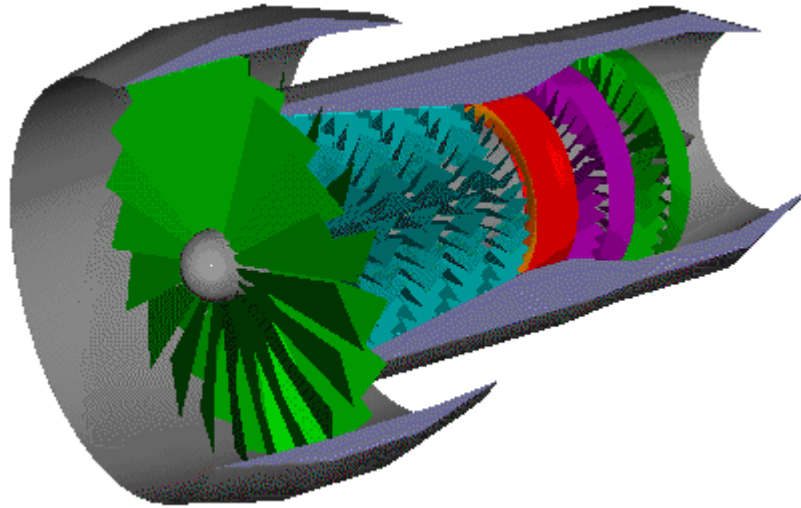


Turbofan Engine

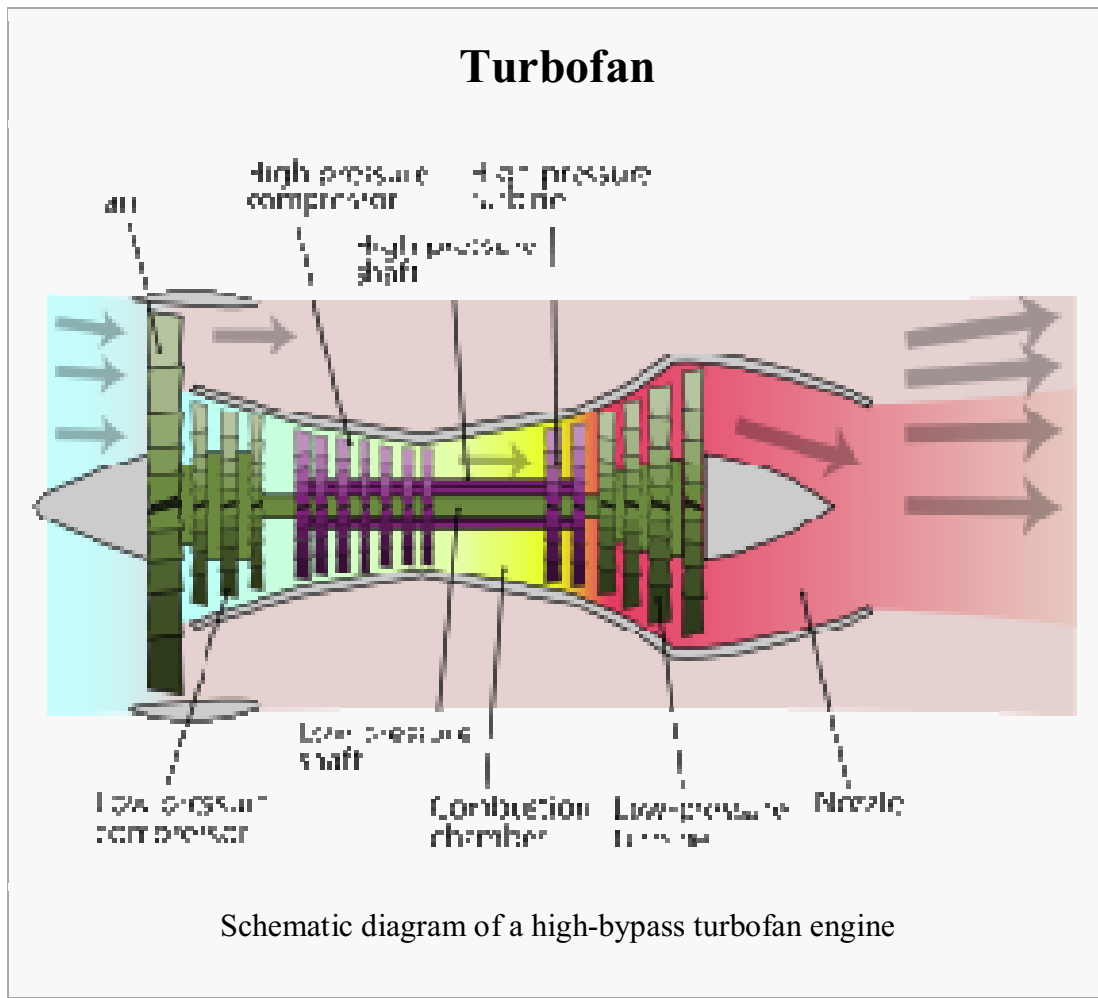


A **turbofan** is a type of [aircraft engine](#) consisting of a [ducted fan](#) which is powered by a [gas turbine](#). Part of the airstream from the ducted fan passes through the gas turbine core, providing oxygen to burn fuel to create power. However, most of the air flow bypasses the engine core, and is accelerated by the fan blades in much the same manner as a [propeller](#). The combination of thrust produced from the fan and the exhaust from the core is a more efficient process than other [jet engine](#) designs, resulting in a comparatively low [specific fuel consumption](#).^[1]

A few designs work slightly differently and have the fan blades as a radial extension of an aft-mounted low-pressure turbine unit.

Turbofans have a net exhaust speed that is much lower than a [turbojet](#). This makes them much more efficient at subsonic speeds than turbojets, and somewhat more efficient at supersonic speeds up to roughly [Mach](#) 1.6, but have also been found to be efficient when used with continuous [afterburner](#) at Mach 3 and above.

All of the jet engines used in currently manufactured commercial jet aircraft are turbofans. They are used commercially mainly because they are highly efficient and relatively quiet in operation. Turbofans are also used in many military jet aircraft.



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To move an [airplane](#) through the air, [thrust](#) is generated by some kind of [propulsion system](#). Most modern airliners use **turbofan** engines because of their high thrust and good [fuel efficiency](#). On this page, we will discuss some of the fundamentals of turbofan engines.

A turbofan engine is the most modern variation of the basic [gas turbine](#) engine. As with other gas turbines, there is a [core engine](#), whose [parts](#) and operation are discussed on a separate page. In the turbofan engine, the core engine is surrounded by a fan in the front and an additional turbine at the rear. The fan and fan turbine are composed of many blades, like the core [compressor](#) and core [turbine](#), and are connected to an additional shaft. All of this additional **turbomachinery** is colored green on the [schematic](#). As with the core compressor and turbine, some of the fan blades turn with the shaft and some blades remain stationary. The fan shaft passes through the core shaft for mechanical reasons. This type of arrangement is called a **two spool** engine (one "spool" for the fan, one "spool" for the core.) Some advanced engines have additional spools for even higher efficiency.

How does a turbofan engine work?

The incoming air is captured by the engine [inlet](#). Some of the incoming air passes through the fan and continues on into the core compressor and then the [burner](#), where it is mixed with fuel and [combustion](#) occurs. The hot exhaust passes through the core and fan turbines and then out the [nozzle](#), as in a

basic [turbojet](#). The rest of the incoming air passes through the fan and **bypasses**, or goes around the engine, just like the air through a [propeller](#). The air that goes through the fan has a velocity that is slightly increased from free stream. So a turbofan gets some of its thrust from the core and some of its thrust from the fan. The ratio of the air that goes around the engine to the air that goes through the core is called the **bypass ratio**.

Because the fuel flow rate for the core is changed only a small amount by the addition of the fan, a turbofan generates more thrust for nearly the same amount of fuel used by the core. This means that a turbofan is very fuel efficient. In fact, high bypass ratio turbofans are nearly as fuel efficient as [turboprops](#). Because the fan is enclosed by the inlet and is composed of many blades, it can operate efficiently at higher speeds than a simple propeller. That is why turbofans are found on high speed transports and propellers are used on low speed transports. Low bypass ratio turbofans are still more fuel efficient than basic turbojets. Many modern fighter planes actually use low bypass ratio turbofans equipped with [afterburners](#). They can then cruise efficiently but still have high thrust when dogfighting. Even though the fighter plane can fly much faster than the speed of sound, the air going into the engine must travel less than the speed of sound for high efficiency. Therefore, the airplane inlet slows the air down from supersonic speeds.

Early turbofans

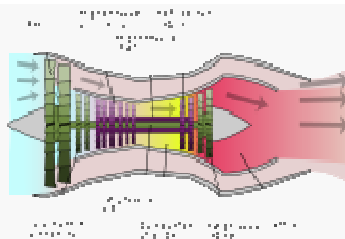
Early turbojet engines were very fuel-inefficient, as their overall pressure ratio and turbine inlet temperature were severely limited by the technology available at the time. The very first running turbofan was the German [Daimler-Benz DB 670](#) (aka 109-007) which was operated on its testbed on April 1, 1943. The engine was abandoned later while the war went on and problems could not be solved. The British wartime [Metrovick F.2](#) axial flow jet was given a fan to create the first British turbofan.

Improved materials, and the introduction of twin compressors such as in the [Pratt & Whitney JT3C](#) engine, increased the overall pressure ratio and thus the [thermodynamic](#) efficiency of engines, but they also led to a poor propulsive efficiency, as pure turbojets have a high specific thrust/high velocity exhaust better suited to supersonic flight.

The original **low-bypass turbofan** engines were designed to improve propulsive efficiency by reducing the exhaust velocity to a value closer to that of the aircraft. The [Rolls-Royce Conway](#), the first production turbofan, had a bypass ratio of 0.3, similar to the modern [General Electric F404](#) fighter engine. Civilian turbofan engines of the 1960s, such as the [Pratt & Whitney JT8D](#) and the [Rolls-Royce Spey](#) had bypass ratios closer to 1, but were not dissimilar to their military equivalents.

The unusual General Electric [CF700](#) turbofan engine was developed as an aft-fan engine with a 2.0 bypass ratio. This was derived from the [T-38 Talon](#) and the [Learjet](#) General Electric [J85/CJ610](#) turbojet (2,850 lbf or 12,650 N) to power the larger Rockwell Sabreliner 75/80 model aircraft, as well as the Dassault Falcon 20 with about a 50% increase in thrust (4,200 lbf or 18,700 N). The CF700 was the first small turbofan in the world to be certified by the Federal Aviation Administration ([FAA](#)). There are now over 400 CF700 aircraft in operation around the world, with an experience base of over 10 million service hours. The CF700 turbofan engine was also used to train Moon-bound astronauts in [Project Apollo](#) as the powerplant for the [Lunar Landing Research Vehicle](#).

Low bypass turbofans



Schematic diagram illustrating a 2-spool, low-bypass turbofan engine with a mixed exhaust, showing the low-pressure (green) and high-pressure (purple) spools. The fan (and booster stages) are driven by the low-pressure turbine, whereas the high-pressure compressor is powered by the high-pressure turbine

A high specific thrust/low bypass ratio turbofan normally has a multi-stage fan, developing a relatively high pressure ratio and, thus, yielding a high (mixed or cold) exhaust velocity. The core airflow needs to be large enough to give

sufficient [core power](#) to drive the fan. A smaller core flow/higher bypass ratio cycle can be achieved by raising the (HP) turbine rotor inlet temperature.

Imagine a retrofit situation where a new low bypass ratio, mixed exhaust, turbofan is replacing an old turbojet, in a particular military application. Say the new engine is to have the same airflow and net thrust (i.e. same specific thrust) as the one it is replacing. A bypass flow can only be introduced if the turbine inlet temperature is allowed to increase, to compensate for a correspondingly smaller core flow.

Improvements in turbine cooling/material technology would facilitate the use of a higher turbine inlet temperature, despite increases in cooling air temperature, resulting from a probable increase in overall pressure ratio.

Efficiently done, the resulting turbofan would probably operate at a higher nozzle pressure ratio than the turbojet, but with a lower exhaust temperature to retain net thrust. Since the temperature rise across the whole engine (intake to nozzle) would be lower, the (dry power) fuel flow would also be reduced, resulting in a better [specific fuel consumption](#) (SFC).

A few low-bypass ratio military turbofans (e.g. F404) have Variable Inlet Guide Vanes, with piano-style hinges, to direct air onto the first rotor stage. This improves the fan surge margin (see [compressor map](#)) in the mid-flow range. The swing wing [F-111](#) achieved a very high range / payload capability by pioneering the use of this engine, and it was also the heart of the famous [F-14 Tomcat](#) air superiority fighter which used the same engines in a smaller, more agile airframe to achieve efficient cruise and Mach 2 speed.

Afterburning turbofans

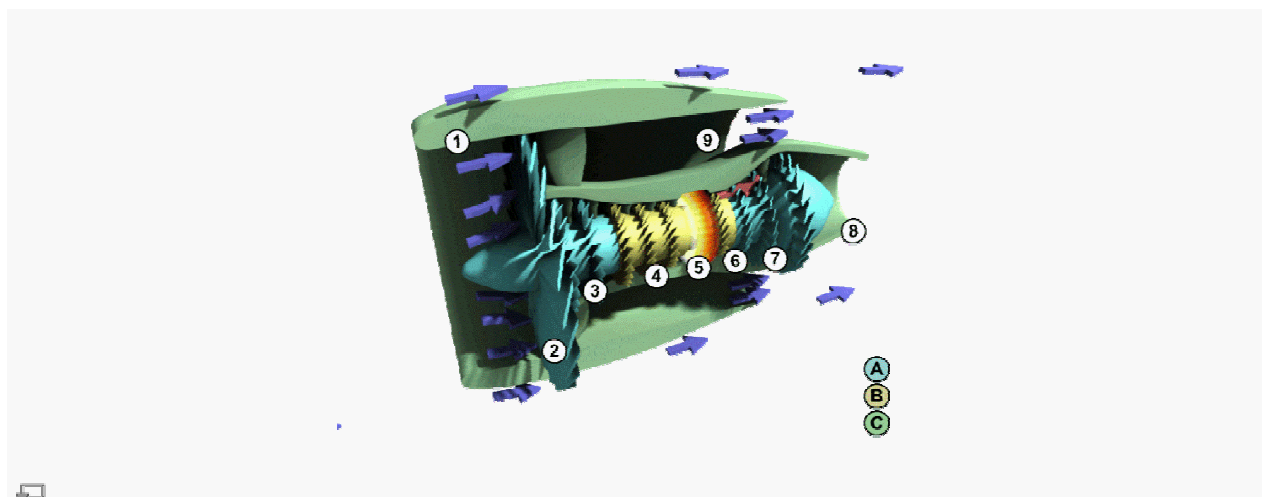
Since the 1970s, most [jet fighter](#) engines have been low/medium bypass turbofans with a mixed exhaust, [afterburner](#) and variable area final nozzle. An afterburner is a combustor located downstream of the turbine blades and directly upstream of the nozzle, which burns fuel from afterburner-specific fuel injectors. When lit, prodigious amounts of fuel are burnt in the afterburner, raising the temperature of exhaust gases by a significant amount, resulting in a higher exhaust velocity/engine specific thrust. The variable geometry nozzle must open to a larger throat area to accommodate the extra volume flow when the afterburner is lit. Afterburning is often designed to give a significant thrust boost for take off, transonic acceleration and combat maneuvers, but is very fuel intensive. Consequently afterburning can

only be used for short portions of a mission. However the Mach 3 [SR-71](#) was designed for continuous operation and to be efficient with the afterburner lit.

Unlike the main combustor, where the downstream turbine blades must not be damaged by high temperatures, an afterburner can operate at the ideal maximum (stoichiometric) temperature (i.e. about 2100K/3780Ra/3320F). At a fixed total applied fuel:air ratio, the total fuel flow for a given fan airflow will be the same, regardless of the dry specific thrust of the engine. However, a high specific thrust turbofan will, by definition, have a higher nozzle pressure ratio, resulting in a higher afterburning net thrust and, therefore, a lower afterburning specific fuel consumption. However, high specific thrust engines have a high dry SFC. The situation is reversed for a medium specific thrust afterburning turbofan: i.e. poor afterburning SFC/good dry SFC. The former engine is suitable for a combat aircraft which must remain in afterburning combat for a fairly long period, but only has to fight fairly close to the airfield (e.g. cross border skirmishes) The latter engine is better for an aircraft that has to fly some distance, or loiter for a long time, before going into combat. However, the pilot can only afford to stay in afterburning for a short period, before his/her fuel reserves become dangerously low.

Modern low-bypass military turbofans include the [Pratt & Whitney F119](#), the [Eurojet EJ200](#) and the [General Electric F110](#) and [F414](#), all of which feature a mixed exhaust, afterburner and variable area propelling nozzle. Non-afterburning engines include the [Rolls-Royce/Turbomeca Adour](#) (afterburning in the [SEPECAT Jaguar](#)) and the unmixed, [vectored thrust](#), [Rolls-Royce Pegasus](#).

High-bypass turbofan engines



- .
- A. *Low pressure spool*
- B. *High pressure spool*
- C. *Stationary components*
 - 1. *Nacelle*
 - 2. *Fan*
 - 3. *Low pressure compressor*
 - 4. *High pressure compressor*
 - 5. *Combustion chamber*
 - 6. *High pressure turbine*
 - 7. *Low pressure turbine*
 - 8. *Core nozzle*
 - 9. *Fan nozzle*



Schematic diagram illustrating a 2-spool, high-bypass turbofan engine with an unmixed exhaust. Again, the fan (and booster stages) are driven by the low-pressure turbine, but more stages are required. A mixed exhaust is often employed nowadays

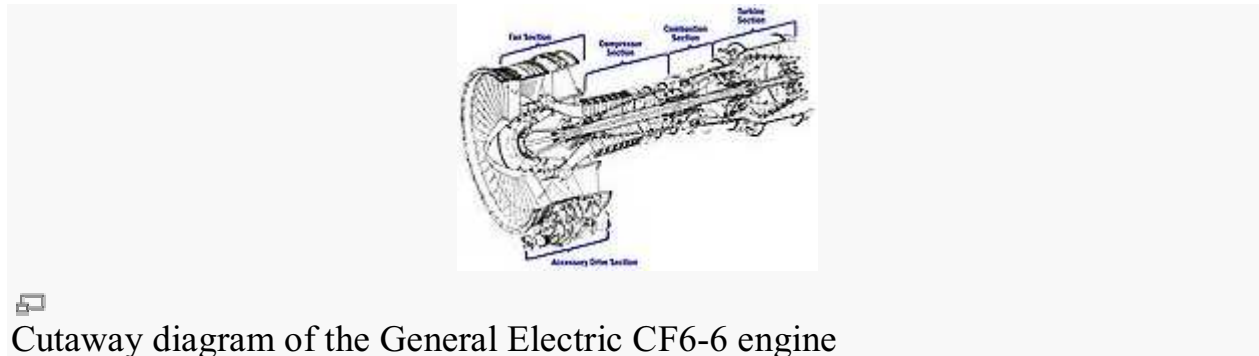
The low specific thrust/high bypass ratio turbofans used in today's civil jetliners (and some military transport aircraft) evolved from the high specific thrust/low bypass ratio turbofans used in such aircraft back in the 1960s.

Low specific thrust is achieved by replacing the multi-stage fan with a single stage unit. Unlike some military engines, modern civil turbofans do not have any stationary inlet guide vanes in front of the fan rotor. The fan is scaled to achieve the desired net thrust.

The core (or gas generator) of the engine must generate sufficient Core Power to at least drive the fan at its design flow and pressure ratio. Through improvements in turbine cooling/material technology, a higher (HP) turbine rotor inlet temperature can be used, thus facilitating a smaller (and lighter) core and (potentially) improving the core thermal efficiency. Reducing the core mass flow tends to increase the load on the LP turbine, so this unit may require additional stages to reduce the average stage loading and to maintain LP turbine efficiency. Reducing core flow also increases bypass ratio (5:1, or more, is now common).

Further improvements in core thermal efficiency can be achieved by raising the overall pressure ratio of the core. Improved blade aerodynamics reduces the

number of extra compressor stages required. With multiple compressors (i.e. LPC, IPC, HPC) dramatic increases in overall pressure ratio have become possible. Variable geometry (i.e. [stators](#)) enable high pressure ratio compressors to work surge-free at all throttle settings.



The first high-bypass turbofan engine was the [General Electric TF39](#), designed in mid 1960s to power the [Lockheed C-5 Galaxy](#) military transport aircraft. The civil [General Electric CF6](#) engine used a derived design. Other high-bypass turbofans are the [Pratt & Whitney JT9D](#), the three-shaft [Rolls-Royce RB211](#) and the [CFM International CFM56](#). More recent large high-bypass turbofans include the [Pratt & Whitney PW4000](#), the three-shaft [Rolls-Royce Trent](#), the [General Electric GE90/GENx](#) and the [GP7000](#), produced jointly by GE and P&W.

High-bypass turbofan engines are generally quieter than the earlier low bypass ratio civil engines. This is not so much due to the higher bypass ratio, as to the use of a low pressure ratio, single stage, fan, which significantly reduces specific thrust and, thereby, jet velocity. The combination of a higher overall pressure ratio and turbine inlet temperature improves thermal efficiency. This, together with a lower

How does an axial compressor work?

The details are quite complex because the blade geometries and the resulting flows are three dimensional, unsteady, and can have important [viscous and compressibility](#) effects. Each blade on a rotor or stator produces a pressure variation much like the [airfoil](#) of a spinning [propeller](#). But unlike a propeller blade, the blades of an axial compressor are close to one another, which seriously alters the flow around each blade. Compressor blades continuously pass through the wakes of upstream blades that introduce unsteady flow variations. Compressor designers must rely on [wind tunnel](#) testing and sophisticated [computational](#)

[models](#) to determine the performance of an axial compressor. The performance is characterized by the pressure ratio across the compressor **CPR**, the rotational speed of the shaft necessary to produce the pressure increase, and an efficiency factor that indicates how much additional work is required relative to an ideal compressor.

Common types

There are two types of jet engine that are seen commonly today, the turbofan which is used on almost all commercial airliners, and [rocket engines](#) which are used for [spaceflight](#) and other terrestrial uses such as ejector seats, flares, fireworks etc.

[\[edit\]](#) **Turbofan engines**

Main article: [Turbofan](#)

Most modern jet engines are actually turbofans, where the low pressure compressor acts as a fan, supplying supercharged air not only to the engine core, but to a bypass duct. The bypass airflow either passes to a separate 'cold nozzle' or mixes with low pressure turbine exhaust gases, before expanding through a 'mixed flow nozzle'.

Turbofans are used for airliners because they give an exhaust speed that is better matched for subsonic airliners, at airliners flight speed conventional turbojet engines generate an exhaust that ends up travelling very fast backwards, and this wastes energy. By emitting the exhaust so that it ends up travelling more slowly, better fuel consumption is achieved as well as higher thrust at low speeds. In addition, the lower exhaust speed gives much lower noise.

In the 1960s there was little difference between civil and military jet engines, apart from the use of [afterburning](#) in some (supersonic) applications. Civil turbofans today have a low exhaust speed (low [specific thrust](#) -net thrust divided by airflow) to keep jet noise to a minimum and to improve fuel efficiency. Consequently the [bypass ratio](#) (bypass flow divided by core flow) is relatively high (ratios from 4:1 up to 8:1 are common). Only a single fan stage is required, because a low specific thrust implies a low fan pressure ratio.

Today's military turbofans, however, have a relatively high specific thrust, to maximize the thrust for a given frontal area, jet noise being of less concern in military uses relative to civil uses. Multistage fans are normally needed to reach

the relatively high fan pressure ratio needed for high specific thrust. Although high turbine inlet temperatures are often employed, the bypass ratio tends to be low, usually significantly less than 2.0.